

Malware Categories, Detection, and Prevevention in OS

1st Zahra Subhi Al Ubaid
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

2nd Sarah Hamdan Alqurashi
College of Computer Science and
Information Technololgy
Imam Abdulrahman bin Faisal
University

3rd Rawan Sudqee AlSaffar
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

4th Haifa Abdullah Alhashim
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

5th Norah Saud Alsubaie
College of Computer Science and
Information Technology
Imam Abdulrahman bin Faisal
University

Abstract— Modern operating systems face increasing cybersecurity threats due to the rapid evolution of malware such as ransomware, rootkits, Trojans, worms, and advanced persistent threats (APTs). These threats exploit operating system vulnerabilities to gain unauthorized access, steal sensitive information, disrupt services, and bypass traditional security mechanisms. This study examines different malware categories, detection techniques, and prevention mechanisms used to protect operating systems against modern cyber threats. The research reviews malware detection approaches including signature-based detection, behavior-based detection, intrusion detection systems (IDS), and artificial intelligence (AI)-driven techniques such as machine learning and deep learning models. In addition, the study analyzes cryptographic protection methods including encryption, hashing, digital signatures, secure communication protocols, and file system encryption in enhancing operating system security. The study also discusses limitations of existing malware detection systems, particularly challenges related to real-time detection, computational overhead, explainability, and zero-day malware attacks. Based on the literature review, the research proposes a lightweight and adaptive security framework that combines AI-based detection, behavior monitoring, and cryptographic protection to improve malware prevention and operating system security.

Keywords—Malware Detection, Operating System Security, Artificial Intelligence, Cybersecurity, Malware Prevention

1. INTRODUCTION

With the advent of advanced digital technology and interconnection among various computing platforms, there has been an increase in the reliance of operating systems on the part of individuals, organizations, governments, and educational institutions. Since operating systems control system resources, user data, application programs, and computer networks, they are often the main target of cyberattacks. One of the most important issues regarding cybersecurity that poses great risks to modern operating systems is malware. Malware represents malicious software programs developed specifically to cause destruction within computers, hamper operations, extract sensitive data, and penetrate computer systems. Malware can take various forms

such as viruses, worms, Trojan horses, spyware, ransomware, adware, and botnets; these software programs use different ways to exploit system weaknesses. Malware has presented a difficult challenge in system protection because hackers keep inventing new tactics in order to avoid standard defense methods against them.

The growing popularity of Internet services, email, cloud-based computing, portable media drives, and file sharing programs has raised the risks for malware infections on operating systems. Malware can propagate through emails, email attachments, downloads from suspicious sources, websites containing malware, wireless networks lacking security measures, and malicious applications. Once installed, malware may cause slow computer performance, damage computer files, consume network bandwidth, monitor the user's activities, encrypt sensitive data, or allow the hacker to remotely access and control the infected machine. Such an environment entails numerous cybersecurity threats that might result in monetary loss, operational disruption, data theft, and privacy violation.

The reason for conducting this research is the increasing requirement to have knowledge about the classification of malware, their operating principles, and the protective measures that can be applied in identifying and preventing malware attacks in an operating system environment. Despite the existence of many types of security tools, including antivirus programs, intrusion detection and prevention systems, firewalls, behavior analysis, and sandboxing, malware has been able to quickly mutate and develop new ways to bypass the security measures.

The current project will concentrate on the classification of malware, detection and prevention methods in operating systems. This work will study various types of malware along with the influence on computer security; it will also examine methods of infection and the security challenges that accompany malware attacks. Furthermore, this project will study various techniques of malware detection and prevention aimed at reducing the danger and increasing cyber security measures. The format of the report is divided into several chapters. The first chapter will discuss the introduction and include the statement of the topic and problems, the aim and objectives of the research, as well as

motivation behind it. The second chapter will provide the literature review and cover the previous studies on malware detection and prevention.

A. Problem Statement

Modern operating systems face increasing cybersecurity threats due to the rapid evolution of malware such as ransomware, rootkits, Trojans, and worms. These threats exploit system vulnerabilities to gain unauthorized access, steal sensitive information, and bypass traditional security mechanisms. Although existing malware detection techniques such as signature-based and behavior-based detection provide protection against known threats, they often struggle to detect zero-day attacks and advanced malware in real time. In addition, current systems face challenges related to computational overhead, false positives, and scalability. Therefore, there is a need for a lightweight and adaptive security framework that combines AI-based detection, behavior monitoring, and cryptographic protection to improve malware prevention in modern operating systems.

B. research methodology and structure of the research

This research follows a qualitative study approach based on reviewing and analyzing recent studies related to malware categories, detection techniques, and prevention mechanisms in operating systems. The research examines different malware detection methods including signature-based detection, behavior-based detection, intrusion detection systems (IDS), and AI-driven techniques such as machine learning and deep learning. In addition, the study analyzes cryptographic protection methods including encryption, hashing, and secure communication protocols. Based on the literature review and comparative analysis, the research proposes a lightweight and adaptive security framework to improve malware prevention in operating systems.

The structure of this research begins with the introduction and background sections, followed by the literature review and gap analysis. The proposed solution section presents the suggested framework, while the discussion and analysis section compares the proposed approach with previous studies. Finally, the research ends with findings, future work, and conclusion.

2. BACKGROUND

Operating systems are essential components of modern computing systems because they manage hardware resources, applications, memory, and network communication. Due to their importance, operating systems are common targets for malware attacks such as ransomware, rootkits, Trojans, and worms. These threats exploit system vulnerabilities to gain unauthorized access, steal sensitive information, and disrupt system operations. Traditional malware detection methods, especially signature-based techniques, are effective against known malware but often

fail to detect new and evolving threats such as zero-day attacks.

To improve operating system security, modern cybersecurity solutions use behavior-based detection, artificial intelligence, and machine learning techniques to identify suspicious activities in real time. In addition, cryptographic protection methods such as AES, RSA, secure boot, and TLS/SSL protocols help protect data confidentiality, integrity, and secure communication. Despite these advancements, challenges such as computational overhead, scalability, and real-time malware detection still remain, making malware prevention an important research area in cybersecurity.

3. LITERATURE REVIEW

a. Ransomware Attacks and Defense Mechanisms in Operating Systems

A comprehensive study by Oz et al. (2022) examined ransomware attacks, their evolution, and the defense mechanisms employed to counter them in modern operating systems. The study explained that ransomware is a category of malware that encrypts victims' files or locks system access, demanding a ransom payment for data recovery. The authors traced the evolution of ransomware from early screen-locking variants such as the AIDS Trojan to highly sophisticated modern families including WannaCry, NotPetya, REvil, and LockBit, which exploit operating system vulnerabilities and propagate across networks autonomously. The study highlighted that ransomware typically operates in multiple stages, including initial access via phishing or vulnerability exploitation, privilege escalation, lateral movement across the network, and finally encryption of critical files using strong asymmetric cryptographic schemes such as RSA and AES. The research also analyzed detection and prevention strategies, including behavioral monitoring for suspicious file encryption activity, network traffic analysis, honeypot-based detection, and backup-based recovery mechanisms. The findings showed that no single defense mechanism is sufficient against modern ransomware; rather, a layered security approach combining real-time behavioral monitoring, cryptographic integrity verification, and regular system backups is necessary to mitigate ransomware risks effectively in operating system environments [1].

b. Trojan Horse and Worm Malware: Characteristics and Detection

Sharma et al. (2023) conducted a systematic review of Trojan horse and worm malware, examining their characteristics, propagation mechanisms, and detection techniques in modern computing environments. The study explained that Trojan horses are malicious programs that disguise themselves as legitimate software to deceive users into executing them, thereby granting attackers unauthorized access to the infected operating system. Unlike viruses and worms, Trojans do not self-replicate but instead create backdoors that allow remote attackers to steal sensitive data, install additional malware, or gain administrative control

over the infected system. The authors further examined worm malware, which propagates autonomously across networks by exploiting operating system and application vulnerabilities without requiring user interaction. The study highlighted that worms can cause significant damage by consuming network bandwidth, disrupting system services, and delivering payloads such as ransomware, spyware, or Distributed Denial of Service (DDoS) agents. The research analyzed various detection techniques applicable to both Trojans and worms, including signature-based detection, network traffic anomaly analysis, static code analysis, and sandboxing-based behavioral analysis. The study concluded that hybrid detection approaches combining static and dynamic analysis provide more accurate and robust identification of Trojan and worm threats compared to relying solely on signature-based methods [2].

c. Advanced Persistent Threats (APTs): Techniques and Detection Strategies

Alshamrani et al. (2019) presented a comprehensive survey of Advanced Persistent Threats (APTs), examining their attack lifecycle, techniques, and detection strategies in enterprise and operating system environments. The study defined APTs as sophisticated, long-term cyberattacks typically carried out by well-resourced adversaries, including nation-state actors and organized cybercriminal groups, with the objective of stealing sensitive data, conducting espionage, or disrupting critical infrastructure. The authors described the APT attack lifecycle, which consists of multiple stages: initial reconnaissance, initial compromise through spear-phishing or zero-day exploits, establishment of persistent backdoors within the operating system, lateral movement to high-value targets, data exfiltration, and covering of tracks to evade detection. The study analyzed several high-profile APT campaigns, including APT28, APT29, and Equation Group, demonstrating how APTs leverage operating system-level techniques such as rootkit installation, process injection, and living-off-the-land attacks using legitimate system tools to maintain long-term persistence while evading conventional security mechanisms. The research reviewed APT detection strategies, including network traffic analysis, threat intelligence sharing, honeypot-based deception systems, machine learning-based behavioral anomaly detection, and Security Information and Event Management (SIEM) systems. The authors concluded that detecting APTs requires continuous monitoring, cross-layer correlation of security events, and integration of threat intelligence to identify the subtle indicators of compromise that APTs leave within operating system environments [3].

d. Signature-Based Malware Detection: Methods, Limitations, and Enhancements

Aslan and Samet (2020) conducted a comprehensive review of signature-based malware detection methods, their limitations, and proposed enhancements for improving detection accuracy in modern operating systems. The study explained that signature-based detection is the most widely deployed malware detection approach, operating by comparing file or code patterns against a database of known malware signatures to identify malicious software. The

authors reviewed the technical implementation of signature-based detection across major antivirus platforms and operating system security modules, explaining that effectiveness relies heavily on the currency of signature databases and the ability to match exact or near-exact code patterns of known malware variants. The study identified several critical limitations of signature-based detection, including inability to detect zero-day malware for which no signature exists, susceptibility to obfuscation techniques such as polymorphism and metamorphism where malware modifies its code to evade signature matching, and high false-negative rates against novel malware families. The research further analyzed proposed enhancements to signature-based detection, including the integration of heuristic analysis to identify suspicious code patterns that resemble known malware behavior, machine learning-based signature generation to automatically derive detection rules from large malware datasets, and hybrid detection frameworks that combine signature matching with behavioral monitoring and anomaly detection. The findings demonstrated that enhancing signature-based detection with machine learning and behavioral analysis significantly improves detection accuracy against obfuscated and previously unseen malware, making it a more viable component of comprehensive operating system security frameworks [4].

a. Operating System Vulnerabilities and Security Risks

Vulnerabilities in several operating systems were analyzed: Windows, Linux distributions, macOS, Android, and Windows Server systems, using data from the National Vulnerability Database (NVD) collected between 1999 and 2022. The results indicated that security vulnerabilities continue to affect operating systems; many of which are classified as high or critical severity. The study also found that operating systems may have weaknesses that attackers can use for malware attacks, unauthorized access, privilege escalation, and data breaches. The study also highlighted the importance of regular vulnerability assessment, security updates, and strong security methods to reduce risks in modern operating systems [5].

b. AI-Based Android Malware Detection

Recently, an AI-based framework called MalTutor was suggested to improve Android malware detection by deep neural networks (DNNs). The study pointed out that many of the existing malware detection models have imbalanced datasets, in which some malware families contain significantly more samples than others. This imbalance has a negative impact on the model's ability to accurately detect rare malware. To tackle this problem, the study proposed an uncertainty-guided curriculum learning method which gradually trains the model by first learning from easier samples before moving to more challenging malware samples. The findings revealed that malware detection accuracy, F1 score, and robustness against different

malware families significantly improved, demonstrating the effectiveness of AI techniques in strengthening operating systems and mobile security systems [6].

c. Evolution of Malware Attacks and Defense Mechanisms

A recent review of a study examined the evolution of malware attacks and modern defense mechanisms in cyber security. The research showed how malware changed from old computer viruses to newer cyber threats such as ransomware, mobile malware, and IoT malware. The study showed that cyber-attacks are becoming more advanced by using methods such as phishing, exploiting security weaknesses, stealth methods, and AI-based attacks to bypass traditional security systems. In addition, the review highlighted the importance of using different protection methods, including artificial intelligence, machine learning, anomaly detection, and behavioral analysis, to improve malware detection and system security. The study also identified several challenges and research gaps caused by the fast development of malware threats and the limitations of current defense methods [7].

d. Linux Rootkits and Detection Challenges

A recent study investigated rootkits for Linux and the limitations of existing rootkit detection tools in modern operating systems. The study investigated 21 Linux rootkits and examined the techniques rootkits use to evade detection such as file hiding, process hiding, network connection hiding, privilege escalation, persistence mechanisms, and backdoor creation. The study also tested various popular detection tools including rkhunter, chkrootkit, ClamAV, OSSEC, and LKRG and found that many of the current detection programs were ineffective in detecting advanced rootkits. Moreover, the research highlighted the need for digital forensics and incident response (DFIR) techniques, such as memory forensics, storage forensics, and network forensics to improve rootkit detection. Modern rootkits are still a big threat to Linux systems, due to their advanced stealth and evasion techniques [8].

a. Encryption Standards: DES, AES, and RSA

A comparison and review of encryption methods was conducted by Kumar and Wagh on DES, AES, and RSA encryption standards. This paper compared the symmetric and asymmetric encryption methods in relation to security, efficiency, and application. DES was stated as one of the old encryption methods because of its short key and vulnerability to brute force attack. On the other hand, AES was described as a very secure and efficient encryption technique appropriate for today's system and data security. RSA was mentioned as a popular asymmetric method for establishing

security keys and creating signatures. It was noted that AES is more powerful than RSA in terms of security with speed, but RSA is applied in cases involving authentication and secure transactions [9].

b. Cryptography Key Management Systems

The study by Alqahtani and Alruily reviewed key management systems in cryptography and their significance in providing a secure environment for communication. The review covered various issues related to the creation, distribution, storage, rotation, and revocation of cryptographic keys in different security models. Some of the techniques of key management included centralized, decentralized, and cloud-based models. The main problems identified included scalability, key leakage, security breaches, and compatibility with new developments. The review noted that key management plays a critical role in ensuring confidentiality, integrity, and authentication in contemporary digital systems. Therefore, organizations need to implement scalable and automated key management techniques to enhance security [10].

c. File System Encryption Techniques

Raza, Shahid, and Khan conducted research on various encryption techniques for files in file systems and the difficulties encountered in implementing those techniques in contemporary computers. The paper reviewed various encryption mechanisms used in ensuring data privacy by securing stored information against any form of unauthorized access, including full-disk encryption, file-level encryption, and other hybrid encryption methods. The authors evaluated the aforementioned encryption techniques according to factors such as efficiency, security level, and compatibility with various operating systems. In addition, some of the implementation difficulties associated with file system encryption, which included encryption overheads, key management complexities, difficulties with data recovery, and decreased computer efficiency, were addressed in the paper. The growing need for encrypted storage mechanisms owing to cloud computing is another point that was highlighted in this paper [11].

d. Post-Quantum Cryptography in SSL/TLS Protocols

Zafar and Iqbal introduced an interoperable hybrid approach in order to incorporate code-based post-quantum cryptography into the SSL/TLS framework. The study emphasized the need to boost security in anticipation of future developments in quantum computation, as these may endanger current cryptographies including RSA and elliptic curve cryptography (ECC). The authors suggested a new hybrid cryptography system, which is able to ensure compatibility between the new and existing SSL/TLS structures. The performance of the hybrid approach was analyzed based on three criteria: security, compatibility, and computational effectiveness. According to the findings, it would be possible to boost the security of communications with regard to quantum attacks. Zafar and Iqbal argued that

hybrid post-quantum cryptographic systems were essential for internet security in the future [12].

a. Advanced Intrusion Detection System Techniques and Challenges

Arnob et al. (2025) conducted a comprehensive systematic review of Intrusion Detection Systems (IDS) to examine emerging techniques, optimization strategies, and future research directions for improving cybersecurity in complex environments such as IoT, cloud computing, healthcare, and industrial networks. The authors analyzed advanced IDS approaches based on Machine Learning (ML), Deep Learning (DL), Explainable Artificial Intelligence (XAI), blockchain integration, and quantum computing technologies to improve intrusion detection accuracy, computational efficiency, scalability, and adversarial robustness. The study highlighted the effectiveness of optimization techniques such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Bayesian Optimization (BO), and Neural Architecture Search (NAS), where several IDS models achieved detection accuracies exceeding 98%. Furthermore, the authors discussed advanced anomaly detection frameworks, transfer learning methods for zero-day attack detection, and adversarial defense mechanisms including ensemble adversarial training and multi-layer filtering to strengthen IDS resilience against sophisticated cyber threats. The review also emphasized the importance of Explainable AI frameworks such as SHAP and LIME in improving transparency, trust, and interpretability of IDS decisions for cybersecurity professionals. Additionally, the study explored blockchain-enabled IDS architectures and Quantum Neural Networks (QNN) to enhance secure communication, data integrity, and real-time threat analysis in decentralized environments. Despite these advancements, the authors identified several critical challenges including real-time detection difficulties, class imbalance in datasets, data complexity, computational overhead, scalability issues, and resource constraints in IoT and edge devices. The study concluded that future IDS research should focus on developing lightweight, adaptive, explainable, and energy-efficient security frameworks capable of operating effectively in dynamic and heterogeneous network environments. [13]

b. Malware Detection Accuracy Challenges

Akour et al. (2016) investigated the challenges associated with malware detection accuracy in real-time and web-based environments by designing and evaluating a malware detection application that communicates with multiple public malware scanning repositories through detection APIs. The study explained that malware continues to evolve rapidly in terms of volume, complexity, and malicious behavior, making accurate detection a major cybersecurity challenge. The authors discussed several malware detection approaches including dictionary-based detection, signature-based detection, static analysis, dynamic analysis, and behavior-

based detection techniques. The study highlighted that traditional signature-based and dictionary-based approaches are effective for detecting known malware but often fail against new and sophisticated malware variants that use code obfuscation and behavioral evasion techniques. To address this issue, the researchers developed an application connected to multiple malware detection engines such as Kaspersky, Sophos, AutoShun, StopBadware, and PhishLabs to evaluate collective detection decisions for malicious websites and files. Experimental results showed that different malware scanners frequently produced conflicting detection outcomes, where some scanners classified a link as malicious while others classified the same link as clean or unrated. The authors also analyzed false positives (FP), false negatives (FN), true positives (TP), and true negatives (TN) as important performance measures in malware detection systems, emphasizing that achieving 100% detection accuracy is practically impossible because of the trade-off between detection accuracy and detection speed. Furthermore, the study identified significant issues related to “unrated” decisions, indicating insufficient information within malware repositories for detecting unknown threats. The authors concluded that future malware detection systems should improve collaborative analysis mechanisms, enhance scanner accuracy evaluation, and develop weighted decision-making algorithms to improve the reliability and effectiveness of real-time malware detection systems. [14]

c. AI-Driven Malware Detection and Prevention Techniques in Operating Systems

Adeyemi D.S. (2025) conducted a systematic review on AI-driven malware detection and classification techniques to examine the effectiveness of artificial intelligence in identifying and preventing malware attacks in operating systems and network environments. The study analyzed multiple AI-based approaches, including machine learning, deep learning, and hybrid detection models such as Random Forest, CNN, RNN, LSTM, XGBoost, and blockchain-integrated AI systems. The author explained that traditional signature-based malware detection methods are no longer sufficient against modern threats such as polymorphic and zero-day malware, which led researchers toward AI-enhanced security solutions. The findings revealed that AI-driven malware detection systems achieved very high detection accuracy, with several studies reporting accuracy rates between 97% and 100%, while also improving malware classification, anomaly detection, and real-time threat identification. Furthermore, the review highlighted important malware analysis techniques including static, dynamic, and hybrid analysis methods, where hybrid approaches demonstrated better robustness and adaptability against complex malware behavior. However, the study also identified several challenges associated with AI-driven malware detection systems, including dataset limitations, computational complexity, adversarial attacks, interpretability issues, and real-time deployment constraints. The author concluded that AI-enhanced malware detection systems significantly improve cybersecurity defenses and recommended continuous dataset updates and explainable AI techniques to strengthen future malware prevention frameworks. [15]

d. Behavior-Based Malware Detection Techniques

Asvica et al. (2024) presented a behavior-based malware detection system designed to identify malware through monitoring software behavior rather than relying only on traditional signature-based detection methods. The authors explained that signature-based antivirus systems are ineffective against novel and rapidly evolving malware because new malware variants may not yet have known signatures. To address this limitation, the proposed system focuses on analyzing behaviors such as file system interactions, registry modifications, process activities, system calls, and network communications in real time. The study discussed the integration of anomaly detection algorithms, heuristic rules, machine learning techniques, and network traffic analysis to identify suspicious activities including code injection, privilege escalation, unauthorized network connections, and malicious process chains. Furthermore, the authors proposed combining the behavior-based detection system with existing cybersecurity tools such as firewalls, antivirus software, intrusion detection systems (IDS), and endpoint detection and response (EDR) systems to improve threat intelligence sharing and malware mitigation. The study concluded that behavior-based malware analysis provides a more resilient and proactive approach for detecting unknown malware and zero-day attacks, although the approach may require high computational resources and extensive monitoring capabilities. [16]

e. Data Leakage Protection and Insider Threat Mitigation

Isabel Herrera Montano, José Javier García Aranda, Juan Ramos Diaz, Sergio Molina Cardín, Isabel de la Torre Díez, and Joel J. P. C. Rodrigues reviewed existing techniques used for data leakage protection and methods designed to address insider threats within organizational systems. The study explained that insider threats caused by employees, collaborators, or authorized users represent one of the major risks to confidential information security. Various protection mechanisms were analyzed, including encryption, hash verification, biometric authentication, virtual file systems, Digital Rights Management (DRM), and hypervisor-based protection techniques. The authors also discussed the integration of machine learning and behavioral monitoring systems to strengthen data leakage prevention and improve the detection of suspicious activities. Furthermore, the study highlighted that intelligent monitoring tools capable of tracking data flow and enforcing predefined security policies can significantly reduce unauthorized disclosure of sensitive information. The review concluded that combining encryption technologies with automated monitoring and machine learning techniques improves the overall effectiveness of organizational data protection systems against insider attacks [17].

f. Malware Detection in Android Operating Systems

Z. D. Patel examined malware detection techniques in Android operating systems and highlighted the increasing challenges caused by rapidly evolving malicious applications. The study reviewed both static and dynamic malware analysis approaches used to identify harmful software within Android environments. Static analysis focused on inspecting application permissions, source code,

API calls, and application structure without executing the program, whereas dynamic analysis concentrated on monitoring runtime behavior such as memory usage, network communication, and system activity during execution. The paper further discussed the application of machine learning algorithms including Support Vector Machine (SVM), Random Forest, Decision Tree, and Neural Networks to improve malware detection accuracy and classification performance. In addition, the study emphasized that anomaly detection, permission analysis, and taint tracking techniques contribute significantly to identifying malicious applications and protecting operating system resources against unauthorized access and data theft. Patel concluded that intelligent malware detection systems are necessary to address the limitations of traditional signature-based security approaches and improve Android operating system security [18].

g. NotPetya Malware Attack and Cybersecurity Response

Steinberg et al. conducted research on the NotPetya cyber-attack and investigated the cybersecurity measures taken as responses to such cyber-attack. The research proved the quick spread of malware attacks via software updates and vulnerabilities, which caused significant disruptions to organizational processes and financial losses. The research focused on the significance of cybersecurity practices such as network segmentation, system isolation, backup, and disaster recovery planning as measures that help to reduce the impacts of massive cyber-attacks. Also, it showed that organizations having robust cybersecurity infrastructure and disaster recovery planning managed to recover from the impact of cyber-attacks in an efficient manner. Steinberg et al. stated that besides conventional approaches for preventing malware attacks, there was also a need for measures related to organizational preparation and recovery to enhance OS security. [19].

h. Malware Family Classification Using Semantic Learning

J. Hao, S. Luo, and L. Pan proposed an enhanced malware family classification framework known as EII-MBS to improve malware analysis through adversarial instruction behavior semantic learning. Malware identification through semantic instruction pattern analysis and the corresponding behavior characteristics of malicious codes formed the core focus of the research. The presented approach improved upon feature extraction and behavior representation stages with an aim at boosting the performance of malware classification.. The researchers evaluated the framework using behavioral analysis and semantic learning techniques, demonstrating improved performance compared to traditional malware classification methods. The study further emphasized the growing importance of artificial intelligence and deep learning approaches in modern malware analysis systems. Hao, Luo, and Pan concluded that semantic learning-based classification frameworks can significantly improve malware detection capabilities and contribute to stronger operating system security mechanisms against evolving cyber threats [20].

A. Gap Analysis

Table I. Malware Categories and Signature-Based Detection

Reference	Aspect Covered	Key Idea	Identified Gap
[1]	Ransomware Attacks	Ransomware uses multi-stage OS exploitation with asymmetric encryption; layered defense combining behavioral monitoring, cryptographic integrity checks, and backups is required.	Existing detection systems struggle to identify ransomware in early encryption stages before significant damage occurs.
[2]	Trojans and Worms	Trojans create backdoors while worms self-propagate via OS vulnerabilities. Hybrid static and dynamic analysis improves detection accuracy.	Signature-based detection remains ineffective against obfuscated Trojans and fast-spreading worm variants.
[3]	Advanced Persistent Threats (APTs)	APTs use multi-stage attacks including rootkit installation and living-off-the-land techniques to achieve long-term OS persistence while evading detection.	Current systems lack the cross-layer correlation and real-time threat intelligence integration needed to identify APT indicators of compromise.
[4]	Signature-Based Detection	Signature matching is effective for known malware but fails against polymorphic and zero-day threats; ML and heuristic enhancements significantly improve detection accuracy.	Signature-based systems cannot adapt to rapidly evolving malware without integration with behavioral and AI-based detection techniques.

Table I summarizes studies related to malware categories and signature-based detection methods. The studies highlight the increasing sophistication of modern malware threats and the limitations of traditional detection approaches. Integrating

behavioral monitoring and AI-based techniques is essential to address these gaps effectively.

Table I. Operating System Vulnerabilities and Malware Evolution

Reference	Aspect Covered	Key Idea	Identified Gap
[5]	OS Vulnerabilities	Analysis of vulnerabilities in Windows, Linux, Android, and macOS systems	Limited proactive defense and prediction methods
[6]	Malware Evolution	Modern malware includes ransomware, APTs, fileless malware, and IoT attacks	Existing defenses struggle against evolving threats

Table I summarize studies related to operating system vulnerabilities and malware evolution. The studies highlight the growing security threats in modern operating systems and the limitations of existing defense mechanisms against evolving malware attacks.

Table II. Malware and Rootkit Detection Techniques

Reference	Research Focus	Main Findings	Research Gaps
[7]	Malware Evolution and Defense Mechanisms	Modern malware uses advanced techniques such as phishing, stealth methods, and AI-assisted attacks to bypass traditional security systems.	Existing defense methods still struggle against rapidly evolving malware threats.
[8]	Linux Rootkits and Detection Challenges	Linux rootkits use stealth techniques such as file and process hiding to evade detection. Many existing	Current detection tools have limited effectiveness against sophisticated rootkits.

		detection tools remain ineffective against advanced rootkits.	
--	--	---	--

Table II summarizes recent studies related to malware evolution and Linux rootkit detection. The studies highlight the increasing sophistication of modern malware attacks and the challenges faced by current detection mechanisms in identifying advanced threats and stealth techniques.

Table III. Encryption Standards and Key Management

Reference	Main Contribution	Research Gap
[9]	AES provides stronger security and performance than DES, while RSA supports secure communication.	No AI-based malware detection integration.
[10]	Key management systems improve confidentiality and authentication.	Limited adaptive security solutions.

Table III The studies concerning encryption standards and key cryptography management are summarized in Table V below. Although there is an emphasis on the need for encryption and authentication, AI-based adaptive security is still underdeveloped.

File System and Post-Quantum Security .Table IV

Reference	Security Technique	Limitation
[11]	File system encryption protects stored data from unauthorized access.	High computational overhead.
[12]	Post-quantum cryptography improves future communication security.	Limited malware detection integration.

Table IV highlights research studies associated with file system encryption and post-quantum cryptography. These research studies contribute to data security and communication security, but real-time malware detection remains difficult.

AI-Driven Malware Detection and Intrusion Detection .Table V Systems

Referenc e	Research Focus	Main Findings	Research Gaps
------------	----------------	---------------	---------------

[13]	Advanced Intrusion Detection System Techniques and Challenges	ML, DL, XAI, blockchain, quantum computing, and optimization algorithms improved IDS accuracy, scalability, and adversarial robustness. Several IDS models achieved detection accuracy above 98%.	The study mainly focused on IDS environments rather than operating system malware detection and lacked a lightweight real-time framework suitable for resource-constrained systems.
[15]	AI-Driven Malware Detection and Prevention Techniques in Operating Systems	AI-driven techniques such as CNN, RNN, LSTM, Random Forest, and XGBoost improved malware detection accuracy, anomaly detection, and zero-day attack identification.	The study was mainly review-based and did not propose a practical lightweight framework for real-time malware detection in operating systems. Computational efficiency and explainability challenges were not fully addressed.

Table V summarizes studies related to AI-driven intrusion detection and malware detection systems. The studies demonstrate that artificial intelligence and deep learning significantly improve malware detection accuracy and adaptability against evolving cyber threats. However, lightweight deployment, explainability, and real-time operating system integration remain major research challenges.

Table VI. Signature-Based and Behaviour-Based Malware Detection Techniques

Referenc e	Research Focus	Main Findings	Research Gaps
[14]	Malware Detection Accuracy Challenges	Signature-based and multi-scanner malware detection systems improved collaborative malware analysis and	Existing approaches struggled with unknown malware, conflicting scanner decisions, false

		detection reliability using API-connected malware repositories.	positives, false negatives, and unrated classifications. Adaptive AI-driven detection mechanisms were limited.
[16]	Behavior-Based Malware Detection Techniques	Real-time behavior monitoring techniques detected suspicious activities such as code injection, privilege escalation, malicious process behavior, and unauthorized network communication. anomaly detection, and zero-day attack identification.	The framework required high computational resources and extensive monitoring capabilities, reducing efficiency in real-time operating system environments. Integration with explainable AI and lightweight optimization methods was limited.

Table VI summarizes studies related to signature-based and behaviour-based malware detection techniques. The studies highlight the importance of real-time behavioural monitoring and collaborative malware analysis in detecting evolving and zero-day malware attacks. Nevertheless, current detection systems still face challenges related to computational overhead, adaptive learning, and efficient real-time deployment in operating system environments.

Table VII .Malware Detection ,Classification ,and Prevention Technique

Reference	Research Focus	Main Findings	Research Gaps
[17]	Data Leakage Protection and Insider Threat Mitigation	,Encryption ,DRM systems biometric ,authentication hypervisor and ,protection machine learning techniques improved confidential data protection and insider threat monitoring within organizational .systems Intelligent monitoring and	The study mainly focused on insider threats and confidential information protection rather than malware ,categories malware ,classification and real-time operating system malware prevention .frameworks Lightweight

		behavioral analysis strengthened data leakage prevention capabilities.	real-time malware protection mechanisms were limited.
[18]	Android Malware Detection Techniques	,Static analysis dynamic ,analysis permission ,analysis anomaly and ,detection machine learning algorithms such ,as SVM ,Random Forest and Neural Networks improved malware detection accuracy in Android operating .systems Intelligent detection methods enhanced identification of malicious applications and suspicious behaviors.	Existing techniques mainly targeted Android environments and faced limitations in detecting unknown malware ,variants adaptive malware and ,behavior efficient real-time operating system .deployment Integration of lightweight AI-driven prevention frameworks remained limited.
[19]	NotPetya Cyberattack Response and Recovery	Network ,segmentation backup recovery ,systems incident response and ,planning system isolation improved organizational resilience against large-scale malware attacks and operational .disruption Cybersecurity preparedness enhanced recovery efficiency following malware incidents.	The study concentrated primarily on recovery and cybersecurity management strategies rather than intelligent malware ,detection behavioral or ,analysis automated malware classification mechanisms for operating system protection.
[20]	Malware Family Classification Using Semantic Learning	Behavioral analysis and adversarial instruction semantic learning improved	The framework focused mainly on malware classification accuracy and

		malware family classification accuracy and enhanced feature extraction for detecting sophisticated malware variants Artificial intelligence and deep learning strengthened malware analysis performance.	lacked integration with comprehensive malware prevention systems lightweight operating system deployment and real-time malware protection optimization techniques.
--	--	--	--

Table VII summarizes studies related to malware detection and malware cyberattack response insider threat mitigation. The reviewed studies demonstrate that classification techniques and semantic learning behavioral analysis machine learning cybersecurity response strategies significantly improve malware real-time. However detection and operating system security lightweight adaptive malware prevention deployment efficiency and unified malware prevention frameworks system integration remain major research challenges in modern operating system environments.

4. PROPOSED SOLUTION

Based on the gap analysis findings from the literature review, this study proposes a Lightweight Adaptive Security Framework (LASF) for operating system malware prevention. The proposed framework integrates three complementary protection layers — AI-based malware detection, real-time behavioral monitoring, and cryptographic data protection — into a unified architecture designed to overcome the key limitations identified in existing systems, specifically the inability to detect zero-day malware, high computational overhead, lack of explainability, and the absence of integrated insider threat mitigation.

A. Framework Architecture Overview

The LASF architecture consists of four main modules that operate in parallel and share threat intelligence through a centralized Security Management Engine (SME): (1) the AI-Based Malware Detection Module, (2) the Real-Time Behavioral Monitoring Module, (3) the Cryptographic Protection Module, and (4) the Explainable AI and Response Module. Each module addresses a specific set of research gaps identified across the reviewed studies [4], [13], [15], [16].

B. Module 1: AI-Based Malware Detection

The first module employs a hybrid AI detection engine combining a Random Forest classifier for static malware feature analysis with a Long Short-Term Memory (LSTM) deep learning model for sequential behavioral pattern recognition. Static features include API call sequences, file permissions, registry key accesses, and code entropy measurements extracted from executable files prior to execution. The Random Forest component analyzes these

static features to detect known malware families and obfuscated variants with high accuracy and low computational cost [15], [4].

The LSTM component processes time-series behavioral sequences collected during program execution, enabling the detection of unknown, polymorphic, and zero-day malware based on behavioral patterns rather than known signatures. This dual-engine design achieves strong detection accuracy — targeting rates above 97% consistent with reviewed AI-based approaches [15] — while maintaining the lightweight computational profile necessary for real-time OS deployment. The hybrid AI engine also incorporates a malware family classification sub-module based on adversarial instruction behavior semantic learning, inspired by the EII-MBS approach reviewed in [20], to support precise malware categorization for forensic and incident response purposes.

C. Module 2: Real-Time Behavioral Monitoring

The second module implements a continuous behavioral monitoring engine that tracks OS-level activities including system calls, file system modifications, registry changes, process creation events, network connection attempts, and privilege escalation requests. An anomaly detection algorithm trained on baseline normal OS activity profiles evaluates all monitored events in real time and assigns anomaly scores to deviations from expected behavior [16], [13].

To address the insider threat gap identified in [17], the behavioral monitoring engine also tracks user-level data flow patterns, application access to sensitive files, and unauthorized data transfer attempts. Behavioral rules derived from the data leakage protection techniques reviewed in [17] are integrated to detect suspicious insider activities such as unauthorized copying of sensitive files, DRM policy violations, and abnormal data access outside normal working patterns. This extension of behavioral monitoring beyond process-level analysis to include user-level data governance distinguishes the proposed framework from existing behavior-based detection systems that focus exclusively on process behavior [16].

The monitoring engine employs a lightweight event filtering mechanism to prioritize high-risk system events, significantly reducing computational overhead compared to full-spectrum monitoring approaches. This addresses the resource consumption limitation identified in the behavior-based detection study reviewed in [16] and the scalability challenge noted in IDS-focused frameworks in [13].

D. Module 3: Cryptographic Protection

The third module integrates a multi-layer cryptographic protection stack to ensure data confidentiality, integrity, and secure communication even under active malware attack conditions. AES-256 encryption is applied to sensitive OS files and user data at rest, drawing on the strong security properties of AES reviewed in [9]. SHA-256 cryptographic hashing is used to generate and maintain integrity checksums for critical system files, executable

binaries, and OS kernel components, enabling immediate detection of unauthorized file modifications characteristic of ransomware [1], rootkit activity [8], and tampering attacks [19].

TLS 1.3 is enforced for all inter-process and network communications to prevent unauthorized data interception and man-in-the-middle attacks, consistent with the secure communication protocol recommendations reviewed in [12]. A centralized key management sub-module based on the principles reviewed in [10] governs the generation, distribution, rotation, and revocation of all cryptographic keys used within the framework, applying automated key rotation policies to minimize the risk of key compromise. Additionally, file system encryption covering both full-disk and file-level protection is applied to sensitive storage areas [11], providing a final layer of data protection in the event that other security layers are bypassed.

E. Module 4: Explainable AI and Automated Response

The fourth module addresses the critical explainability and automated response gaps identified across multiple reviewed studies [13], [15]. An Explainable AI (XAI) component using SHAP (SHapley Additive exPlanations) values is integrated into the AI detection engine to generate human-readable explanations for every malware detection decision. SHAP values identify the specific features — such as suspicious API call sequences, anomalous registry modifications, or unusual network connections — that most strongly contributed to each detection outcome. These explanations are logged and presented to security analysts through a unified security dashboard, improving trust, transparency, and the ability to investigate and respond to detected threats [13].

An automated incident response sub-module is connected to the detection and monitoring outputs of all three previous modules. When a threat is confirmed with high confidence, the response sub-module automatically triggers predefined remediation actions including process termination, network connection blocking, quarantine of suspicious files, and rollback to the most recent verified system state using snapshot-based recovery. For ransomware threats in particular, the response module monitors file encryption rate metrics and triggers emergency backup preservation and network isolation if encryption activity exceeds predefined thresholds, addressing the early-stage ransomware detection gap identified in [1]. Incident reports incorporating SHAP-based explanations, attack timeline reconstruction, and remediation actions are automatically generated to support digital forensic analysis and organizational cybersecurity response planning [19].

F. Framework Integration and Scalability

The four modules share threat intelligence through the centralized Security Management Engine, which correlates detection signals across the AI engine, behavioral monitor, cryptographic integrity checker, and XAI response module to reduce false positives and improve detection confidence for complex multi-stage attacks such as APTs [3]. All detection models are designed using lightweight

architectures suitable for deployment across a range of OS environments, from server-grade systems to resource-constrained endpoint devices, addressing the scalability limitations identified across multiple reviewed studies [13], [16].

The framework supports continuous model updating through an offline retraining pipeline that incorporates newly discovered malware samples and behavioral signatures, ensuring that the AI detection engine remains effective against rapidly evolving malware threats over time [7], [15]. Android OS deployment compatibility is supported through a mobile-optimized variant of the behavioral monitoring and AI detection components, extending the framework's applicability to Android malware detection scenarios reviewed in [6] and [18].

1 DISCUSSION AND ANALYSIS

The proposed Lightweight Adaptive Security Framework (LASF) was conceived to fill some of the major gaps in the previous studies on OS malware prevention and cybersecurity defense mechanisms. The designed framework combines AI, behavioral monitoring, cryptographic security, and explainable AI into a multi-layered framework, going beyond the conventional approach of relying on a single detection method. This integration enhances the security and accuracy of malware detection, bolstering real-time threat response, and the security of the operating system.

The earlier research showed that traditional signature-based malware detection systems are not effective in detecting zero-day malware, polymorphic malware, and obfuscated malicious code, as they rely primarily on known malware signatures [4] and [15]. While signature-based systems are good for detecting known variants of a virus, they are unable to detect new variants of malware. The proposed LASF overcomes this limitation by combining Random Forest-based static analysis with LSTM-based behavioral sequence analysis. This hybrid AI architecture enables the framework to identify known and unknown malware through behavioral pattern recognition rather than relying on malicious signatures stored in databases. Additionally, the shortcomings of previous approaches are addressed by the use of semantic learning techniques inspired by the EII-MBS model [20] for malware family classification, which enhances the adaptability of the framework over the previously mentioned AI-based malware detection methods in [15] and [18]. This improves malware classification and forensic analysis.

In [16], the authors found that behavior-based malware detection systems are more capable of detecting unknown malware than traditional antivirus systems. But such systems tend to be resource-intensive because they are constantly monitoring everything the operating system is doing. In the same way, the studies reviewed in [13] that focused on IDS reported scalability and resource use as some of the main

challenges of real-time cybersecurity systems. The proposed framework overcomes these drawbacks by providing an efficient event filtering mechanism which is used to prioritize high-risk activities and minimize unnecessary monitoring operations. This makes the framework more scalable and better suited for deployment in resource-limited environments without significantly affecting detection performance.

Further, most existing behavior-based malware detection systems focused mainly on process-level operations like network communication, registry changes, and system calls [16]. But there was relatively little discussion of insider threat mitigation and of user-level protection. In [17], the study reviewed the need to combine behavioral monitoring with data leakage prevention systems to mitigate insider attacks and unauthorized access to sensitive data. The proposed framework enhances behavioral analysis with user-level activity monitoring, sensitive file access monitoring, and abnormal data transfer detection based on this gap. This provides a more comprehensive approach than the other malware detection approaches which target only the action of malware.

Cryptographic protection also plays an important role in operating system security, as discussed in several reviewed studies [1], [9], [10], [11], and [12]. Previous studies primarily focused on individual cryptographic methods, like AES encryption, TLS protocols, or file system encryption. The proposed LASF, on the other hand, combines several cryptographic security measures into a single coordinated security layer. Sensitive stored data is protected by AES-256 encryption, file integrity is ensured with the help of the SHA-256 hash function, TLS 1.3 secures the communication channels, and centralized key management provides secure cryptographic operations. This multilayered cryptographic solution provides stronger resistance against ransomware attacks, rootkits, tampering attempts, and unauthorized data access, compared to isolated encryption solutions discussed in previous studies.

Another crucial problem tackled by the proposed framework is the explainability gap that has been highlighted in the studies on AI-based IDS and malware detection [13, 15]. However, many of the AI systems developed for cybersecurity are “black box” systems, and analysts have a hard time understanding why decisions were made for certain detections. The proposed framework includes Explainable AI techniques based on SHAP to get transparent explanations of malware detection and anomaly alert. This helps analysts gain trust, aids in incident investigations, and enables better decision-making within the cybersecurity strategy. Unlike previous studies which suggested explainable AI as future solutions [13], the proposed framework incorporates explainability as a fundamental part of the architecture.

The proposed framework also provides stronger automated incident response capabilities compared to previous studies. Although major research works like [1] and [19] emphasized the need for backup mechanisms, recovery planning, and ransomware mitigation, they were unable to incorporate automated response systems with AI detection and behavioral monitoring engines. LASF integrates all protection modules via a shared Security Management Engine (SME) to provide synchronized threat correlation and automatic remediation operations, including process termination, file quarantine, and network isolation, rollback recovery and emergency backup storage. This integration helps to improve response speed and minimize the possible damage resulting from sophisticated attacks caused by advanced malware or ransomware outbreaks.

In addition, the proposed framework covers scalability and adaptability constraints found in various studies reviewed [7, 13, 16]. Existing malware detection systems face difficulties adapting to the constant changes in malware activity and attack methods. To keep the detection models up to date with modern threats, LASF offers continuous offline retraining with newly collected malware samples and behavioral signatures. Moreover, the framework supports Android, broadening its use beyond traditional desktop operating systems to make it applicable to today's more diverse computing platforms.

In conclusion, the proposed Lightweight Adaptive Security Framework offers a more comprehensive and adaptive security solution than the existing malware prevention frameworks. The framework is designed to fill several gaps in the literature identified during the literature survey, including the following features: AI-based malware detection, lightweight behavioral monitoring, integrated cryptographic protection, explainable AI, insider threat mitigation, and automated response. The proposed design has strong potential to enhance the effectiveness of malware prevention, operating system resilience, and effectiveness of cybersecurity response efforts in modern computing environments.

2 FINDINGS AND FUTURE WORK

From the results of this study, it can be seen that modern operating systems are still subject to significant cybersecurity threats due to the rapid emergence of different kinds of malware attacks, including ransomware, rootkits, Trojans, worms, and zero-day malware. Signature-based malware detection was quite effective when it came to detecting known malware types; however, its potential was rather limited in terms of advanced and new malware attacks. Behavior-based malware detection and other approaches based on artificial intelligence, like machine learning and deep learning were much more accurate and versatile when dealing with suspicious operations and unknown malware attacks. It can also be concluded from the research that cryptographic methods for data protection, including AES, RSA, key management, file system encryption, and SSL/TLS

communication protocols provided effective protection of data integrity, confidentiality, and security in operating systems. Nevertheless, there is a number of problems, which include computational load, false alarms, the explainability problem, and delays in malware detection. In the future, researchers should concentrate their attention on developing effective security solutions combining malware detection through AI tools, behavior detection, and cryptographic protection in order to provide efficient security for future OS versions.

3 CONCLUSION

In summary, the study highlighted the emerging cybersecurity threats targeting modern operating systems because of the fast development of malware attacks, which include ransomware, rootkits, Trojans, worms, and persistent threats. Malware attacks take advantage of flaws within the operating system to obtain unauthorized access, steal confidential information, evade existing security measures, and interrupt operations. This study uncovered various issues in current malware detection solutions, which include inadequate performance in combating zero-day attacks, excessive computational costs, high error rates, limited scalability, and real-time accuracy. Besides, signature-based malware detection was deemed ineffective in fighting modern and constantly evolving malware attacks.

This paper analyzed several contemporary cybersecurity strategies such as behavior-based malware detection, intrusion detection systems, AI approaches, machine learning algorithms, deep learning algorithms, encryption protocols, key management systems, file system encryption, and secure communication protocols. The review of literature and comparative analysis led to the design of a flexible security model that incorporates malware detection through artificial intelligence, behavior monitoring, and cryptographic approaches. The suggested approach is anticipated to enhance malware prevention, detection accuracy, and data confidentiality and integrity. Artificial intelligence can be integrated into modern cybersecurity approaches alongside cryptographic techniques to effectively protect the operating system against present and future cyber risks.

REFERENCES

- [1] H. Oz, A. Aris, A. Levi, and A. S. Uluagac, "A Survey on Ransomware: Evolution, Taxonomy, and Defense Solutions," *ACM Computing Surveys*, vol. 54, no. 11s, pp. 1-37, 2022, doi: 10.1145/3514229.
- [2] A. Sharma, K. Yadav, A. K. Singh, and A. K. Gautam, "A Survey on Trojan Horse and Worm Malware: Characteristics, Detection Techniques, and Future Research Directions," *Journal of Computer Virology and Hacking Techniques*, vol. 19, pp. 45-68, 2023.
- [3] A. Alshamrani, S. Myneni, A. Chowdhary, and D. Huang, "A Survey on Advanced Persistent Threats: Techniques, Solutions, Challenges, and Research Opportunities," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 2, pp. 1851-1877, 2019, doi: 10.1109/COMST.2019.2891891.
- [4] O. A. Aslan and R. Samet, "A Comprehensive Review on Malware Detection Approaches," *IEEE Access*, vol. 8, pp. 6249-6271, 2020, doi: 10.1109/ACCESS.2019.2963724.
- [5] M. Bhurtel and D. B. Rawat, "Unveiling the Landscape of Operating System Vulnerabilities," *Future Internet*, 2023.
- [6] H. Li, X. Cheng, Y. Zhao, G. Xu, G. Xu, and H. Wang, "Understanding Model Weaknesses: A Path to Strengthening DNN-Based Android Malware Detection," *Proceedings of the ACM on Software Engineering*, 2025.
- [7] J. Ferdous, R. Islam, A. Mahboubi, and M. Z. Islam, "A Review of State-of-the-Art Malware Attack Trends and Defense Mechanisms," *IEEE Access*, 2023.
- [8] J. Stühn, J.-N. Hilgert, and M. Lambertz, "The Hidden Threat: Analysis of Linux Rootkit Techniques and Limitations of Current Detection Tools," *Digital Threats: Research and Practice*, 2024.
- [9] R. Kumar and S. D. Wagh, "A Review Paper on DES, AES, RSA Encryption Standards," *IEEE Access*, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/9336800/>
- [10] H. M. H. S. Alqahtani and M. A. F. Alruily, "A comprehensive survey of cryptography key management systems," *Journal of Information Security and Applications*, vol. 78, art. no. 103607, 2023, doi: 10.1016/j.jisa.2023.103607.
- [11] A. Raza, M. Shahid, and M. A. Khan, "A survey on file system encryption techniques and implementation challenges," *Computers & Security*, vol. 120, 2022, doi: 10.1016/j.cose.2022.102785.
- [12] A. Zafar and S. S. Iqbal, "Integrating code-based post-quantum cryptography into SSL/TLS protocols through an interoperable hybrid framework," *Discover Computing*, vol. 28, art. no. 202, 2025, doi: 10.1007/s10791-025-09735-7.
- [13] Arnob, A. K. B., et al., "A comprehensive systematic review of Intrusion detection systems: emerging techniques, challenges, and future research directions," *Journal of Engineering and Cybersecurity*, 2025.
- [14] T. Alsmadi and N. Alqudah, "A Survey on malware detection techniques," 2021 International Conference on Information Technology (ICIT), Amman, Jordan, 2021, pp. 371-376, doi: 10.1109/ICIT52682.2021.9491765.
- [15] Adeyemi, D. (2025) AI-driven malware detection and classification, *European Journal of Computer Science and Information Technology*. Available at: <https://eajournals.org/ejcsit/wp-content/uploads/sites/21/2025/10/AI-Driven-Malware-Detection.pdf>
- [16] J. Asvica, R. S. Dhanusree, S. Dha'shan', S. Lavanya, P. N. Sowbarnika, S. Yaswanthraj et al. (2024) International Journal of Advances in Engineering and Management (IJAEM). Available at: https://ijaem.net/issue_dcp/Behavior%20Based%20Malware%20Detection.pdf
- [17] Isabel Herrera Montano, José Javier García Aranda, Juan Ramos Díaz, Sergio Molina Cardín, Isabel de la Torre Díez, and Joel J. P. C. Rodrigues. 2022. Survey of Techniques on Data Leakage Protection and Methods to address the Insider threat. *Cluster Computing* 25, 6 (Dec 2022), 4289–4302. <https://doi.org/10.1007/s10576-022-03668-2>
- [18] Z. D. Patel, "Malware Detection in Android Operating System," 2018 International Conference on Advances in Computing, Communication Control and Networking (ICACCCN), Greater Noida, India, 2018, pp. 366-370, doi: 10.1109/ICACCCN.2018.8748512.
- [19] Steinberg, S., Stepan, A. and Near", K. (2017) Notpetya: A Columbia University Case Study, School of International and Public Affairs Case Consortium @ Columbia . Available at: <https://www.sipa.columbia.edu/sites/default/files/2022-11/NotPetya%20Final.pdf>
- [20] J. Hao, S. Luo, and L. Pan, "EII-MBS: Malware family classification via enhanced adversarial instruction behavior semantic learning," *Computers & Security*, vol. 122, p. 102905, 2022, doi: 10.1016/j.cose.2022.102905.